PLATFORM SPECIFICATION & ARCHITECTURAL DOSSIER

endorphix

The Luxury Athletic Catalyst & Spontaneous Vitality Engine

> *"The luxury of movement is a gift given to many that shall not be wasted. Running, jumping, and channeling energy throughout the body is the most effective way of getting to know yourself."*
>
> **— endorphix Brand Philosophy**

1. Platform Identity & Design Philosophy

endorphix is a luxury athletic platform engineered for high-intensity athletes, competitors, and discerning movers. It eliminates workout procrastination, choice paralysis, and arbitrary calorie-counting by connecting athletes directly to top-tier verified sports facilities and tracking somatic vitality.

Chalk `#EDEBE6 (Base Canvas)`	**Forest Olive** `#5A6738 (Action Accent)`	**Deep Navy** `#22314E (Heritage Core)`	**Terracotta** `#6F2E1B (Spark / Alert)`
Camel `#A7744B (Realm Highlight)`	**Sand** `#E0D4C1 (Card Surfaces)`	**Sand Dark** `#D5C8B4 (Subtle Borders)`	**Espresso** `#2B1F17 (High-Contrast Text)`

2. Feature Architecture & Core Pillars

DUAL-PATH ENGINE

Discover

Sport-First filtering across 3 Macro Realms (Water, Land, Mountain) and Destination-First multi-day athletic retreat generator with 10/25/50km radius control.

ACTION CATALYST

The Spark

Single-touch spontaneous engine querying 4.5★+ facilities within 50km with live availability, Apple/Google Maps routing, and instant court locking.

VITALITY RECORD

Passport

Mathematical D3/SVG endorphin wave, streak tracking, credential portrait picker, notifications manager, and zero-lock-in JSON data export.

The Three Macro Sport Realms

- WATER **Hydrodynamic conditioning:** Swimming, rowing, open-water swell surfing, sailing.
- LAND **Kinetic mastery & court craft:** Clay/grass tennis, high-cadence running, cycling, padel, athletic gym, boxing, combat arts.
- MOUNTAIN **Alpine endurance:** Vertical rock ascent, bouldering, alpine trail running, high-altitude hiking.

Athletic Valuation Index & Task Metrics

Evaluates cardiovascular load, fast-twitch motor units, agility, strength, and booking frequency across intervals (Today, Week, Month, 6 Months, All Time) to dynamically identify the athlete's most valuable conditioning sport.

3. The Endorphin Bar & 30-Minute Recovery Protocol

Rather than interrupting athletic flow during a training session, endorphix uses a physiological state machine: **30 minutes post-session** (when lactate flushes and central nervous endorphins peak), an in-app alert banner triggers prompt somatic logging.

90 – 100	**Peak Vitality High**	Transcendent neuromuscular flow & euphoric endorphin flush
75 – 89	**High Euphoric Flow**	Elevated central nervous clarity, buoyant somatic lightness
50 – 74	**Active Flow State**	Clean baseline activation, centered cardiovascular cadence
25 – 49	**Grounded & Resetting**	Subdued neuromuscular state, tissue conditioning
1 – 24	**Somatic Depletion**	Cellular replenishment & hydrotherapy recommended

4. Technical Logic & Workflows

```
[Athlete Selects Slot] ──> [Booking: Confirmed] ──> [Session Concluded] │ [D3 Curve Updated] <── [1-100 Somatic Slide] <── [30m Alert Triggered]
```

- **Geolocation & Failover Logic:** Queries PostgreSQL / PostGIS `ST_DWithin` (up to 50km). Automatically fails over to OpenStreetMap (OSM) Overpass API and curated patron data if network or provider quotas fluctuate.
- **Mathematical Curve Interpolation:** Computes cubic bezier splines with control offset ratio `(x_{i+1} - x_i) / 2.5`, rendering dynamic peak scores and closed shaded area polygons via React Native SVG.
- **Universal Responsiveness:** Built-in `useResponsive` hook dynamically recalculates layouts across phones (single column, full-width bottom bar), tablets / iPads (2 columns, centered floating tab capsule), and laptops / desktops (3 columns, centered max-width).
- **Data Sovereignty & Offline-First:** Built-in *Download My Data* JSON generation allows instant export of bookings, somatic logs, and passport metrics with zero vendor lock-in.

5. Complete Technology Stack

Layer	Technology	Role & Responsibility
Framework	React Native 0.76.7 · Expo 52	Universal cross-platform core for iOS, Android, and Web
Routing	Expo Router v4	File-based navigation with modal presentations and typed routing
Language	TypeScript 5.7	Strict domain interfaces, GeoPoints, and type guarantees
Styling	NativeWind v4 · Tailwind CSS 3.4	Utility-first styles with Earthy Luxury token system
State Stores	Zustand v5	Decoupled reactive stores for bookings, auth, location, and discovery
Server State	TanStack React Query v5	Cache invalidation, background polling, and optimistic UI updates
Database	Supabase · PostgreSQL · PostGIS	Spatial indexing, radius queries, and Row-Level Security (RLS)
Vector & Charts	React Native SVG 15.11	Mathematical bezier curves, custom cursors, and visual rings
Motion & Haptics	Moti · Reanimated 3.17 · Expo Haptics	Spring entrances, tactile micro-vibrations, and fluid transitions
Responsive Engine	Custom `useResponsive` + `ResponsiveContainer`	Fluid adaptive layout supporting phone, tablet, iPad, laptop & desktop
App Server	Node.js HTTP Server (`serve-app.js`)	Zero-dependency lightweight SPA server running on port 8081